

Ten-year results of transscleral resection of large uveal melanomas: local tumour control and metastatic rate

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ABSTRACT

Aims To describe the long-term tumour control and metastatic rate after transscleral resection (TSR) of large uveal melanomas in a single-centre study.

Methods The sample included 210 patients with large uveal melanomas. Univariate analysis of local tumour control and metastatic risk by Kaplan—Meier survival curves and log-rank testing. Cox proportional HR analysis with forward and backward selection was used to identify independent prognostic factors in patients submitted to TSR of a large uveal melanoma.

Results A residual tumour was diagnosed in 6% of the patients. The 5- and 10-year local tumour recurrence rates were 24% and 32%, respectively. Older age, a large basal tumour diameter, the lack of adjuvant ruthenium brachytherapy and retinal detachment led to a 2.6, 2.4, 4.4 and 7.8 times higher risk of melanoma recurrence, respectively. The 5- and 10-year metastatic rates were 28% and 44%, respectively, and were statistical significantly affected by extraocular spread, tumour thickness and local tumour recurrence.

Conclusions TSR is an alternative to enucleation for the treatment of large uveal melanomas. Results should improve with better patient selection and more effective methods of adjuvant radiotherapy.

INTRODUCTION

Treatment of uveal melanoma has become a focus of intensive controversy, with debates ranging from whether eye salvaging is justified to whether enucleation might promote melanoma cell dissemination and metastasis.^{1–3} This issue has been resolved by the prospective randomised Collaborative Ocular Melanoma Study (COMS) showing no difference in survival between ‘conservative’ treatment with ¹²⁵I brachytherapy and enucleation.⁴ In the case of large uveal melanomas, eye retention therapy remains a challenging issue with regard to tumour control and treatment-related complications. ¹⁰⁶Ru brachytherapy is insufficient for controlling tumours thicker than 7 mm, and ¹²⁵I brachytherapy for large uveal melanomas increases the incidence of eye-threatening anterior and posterior segment complications.^{5–8} Proton beam radiotherapy offers an alternative approach, but is also associated with eye-threatening complications when used to treat large tumours.⁹ Transscleral resection (TSR) of uveal melanomas is a surgical treatment option that has evolved considerably from when it was first reported in the 1960s by Stallard and Müller.^{10–12} Foulds started performing

local resection irrespective of the state of the fellow eye, unlike Stallard, who reserved local resection only for seeing eyes.¹³ Despite the development of innovative microsurgical techniques and instruments, TSR remains a technically challenging procedure.

This study is a retrospective, non-comparative, consecutive case series of patients with uveal melanomas treated by TSR. Its aim was to evaluate the long-term local tumour control results and metastasis rate after TSR and to identify possible risk factors.

PATIENTS AND METHODS

A total of 210 patients with uveal melanoma, consisting of 57 cases with choroidal melanoma and 153 cases with ciliochoroidal melanoma, were submitted to TSR under systemic hypotension between October 1992 and December 2007. The first 20 patients underwent a TSR without adjuvant radiotherapy to the tumour area. After September 1993, surgery was combined with adjuvant ¹⁰⁶Ru plaque brachytherapy (n=136). After August 2004, adjuvant ¹⁰⁶Ru plaque brachytherapy was replaced by ‘neoadjuvant’ preoperative proton beam radiotherapy. All 54 patients thus treated were excluded from this study, as well as six patients with a follow-up of less than 3 months. A total of 150 patients were included in the final analysis.

Preoperative assessment and eligibility criteria

All patients underwent a complete ophthalmological examination. Tumour thickness, largest basal tumour diameter (LBD), and the distances from the posterior tumour margin to the optic disc (TDO) and fovea (TDF), were measured by ultrasound. The presence or absence of ciliary body infiltration was assessed by ophthalmoscopy and by transillumination.

TSR was considered in patients who were highly motivated to keep their eye, containing a uveal melanoma ineligible for ¹⁰⁶Ru plaque brachytherapy or proton beam irradiation. In particular, all tumours with the following characteristics were eligible for TSR: a minimum tumour thickness of 7.0 mm irrespective of ciliary body or anterior chamber invasion, a posterior margin no closer than 3 mm to the optic disc and/or fovea, and a LBD of <23 mm. The present study also included seven ciliary body melanomas with tumour thickness of less than 7.0 mm (figure 1). No iris melanomas were included in this series.

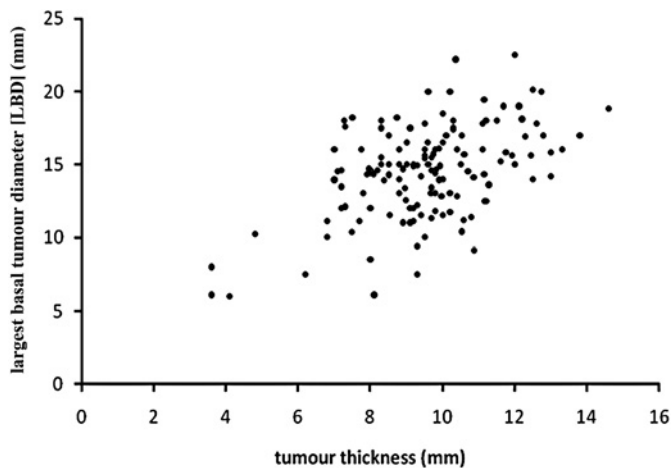


Figure 1 Scatter plot showing largest basal tumour diameter (LBD) over tumour thickness (height) of all tumours.

Table 1 Demographic characteristics of the 150 patients (77 women, 73 men) that were included in the final analysis of the study

Characteristic	Mean	SD	Range
Baseline visual acuity (decimal)	0.4		HMP–1.0
Tumour thickness (mm)	9.4	2.0	2.0–14.6
LBD (mm)	14.5	3.1	6.0–22.5
TDO (mm)	10.8	4.3	4.1–20.8
TDF (mm)	10.4	4.6	1.0–20.8
Follow-up (months)	45.5 (median=37)	31.6	3–142

HMP, hand movement perception; LBD, largest tumour basal diameter; TDF, tumour distance to fovea; TDO, tumour distance to optic disc.

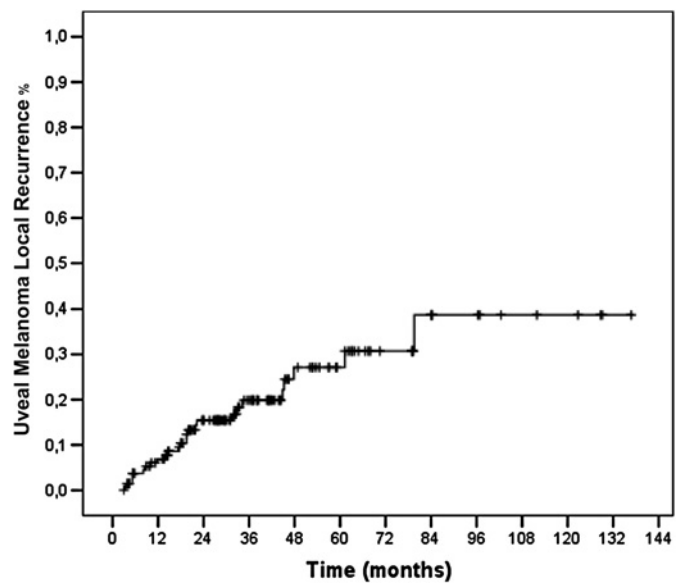


Figure 2 Kaplan–Meier curve showing the overall incidence of uveal melanoma recurrence after transscleral resection (TSR).

Only patients with a preoperative visual acuity ≥ 0.05 , irrespective of the status of the fellow eye, were offered this procedure.

All patients underwent a medical check-up including abdominal ultrasound to exclude liver metastasis, a general neurological examination, an electroencephalogram (EEG), a Doppler ultrasound examination of the external carotid arteries, an orbital MRI, 24-h blood pressure monitoring, a 24-h electrocardiogram (ECG) and a stress ECG.

Table 2 Recurrent tumour after transscleral resection

Univariate analysis (log-rank test)				Multiple Cox regression analysis after forward and backward analysis		
Variable	Category	No. of patients with recurrent tumour/total no. of patients	p Value	HR	95% CI for HR	p Value
Sex	Female	10/73	0.19	2.6	1.0 to 6.6	0.046
	Male	16/68				
Age (years)	<50*	8/62	0.09			
	≥50	18/79				
Baseline visual acuity (decimal)	<0.5	17/75	0.15			
	≥0.5	9/66				
Ciliary body infiltration	Yes	18/103	0.78			
	No	8/38				
Tumour thickness (mm)	≤9.5	13/71	0.79			
	>9.5	13/69				
LBD (mm)	<16*	12/92	0.00	2.4	1.1 to 5.6	0.035
	≥16	14/43				
TDO (mm)	<10	11/72	0.24			
	≥10	15/61				
TDF (mm)	<10	9/58	0.46			
	≥10	10/43				
¹⁰⁶ Ru brachytherapy	Yes*	19/124	0.04	4.5	1.8 to 11.1	0.001
	No	7/17				
Histological type	Spindle cell	9/58	0.25			
	Mixed or epithelioid cell	17/83				
Exudative retinal detachment	Yes	24/97	0.01	7.8	1.0 to 59.3	0.046
	No*	1/34				

*Reference category in Cox regression analysis.

LBD, largest tumour basal diameter; TDF, tumour distance to fovea; TDO, tumour distance to optic disc.

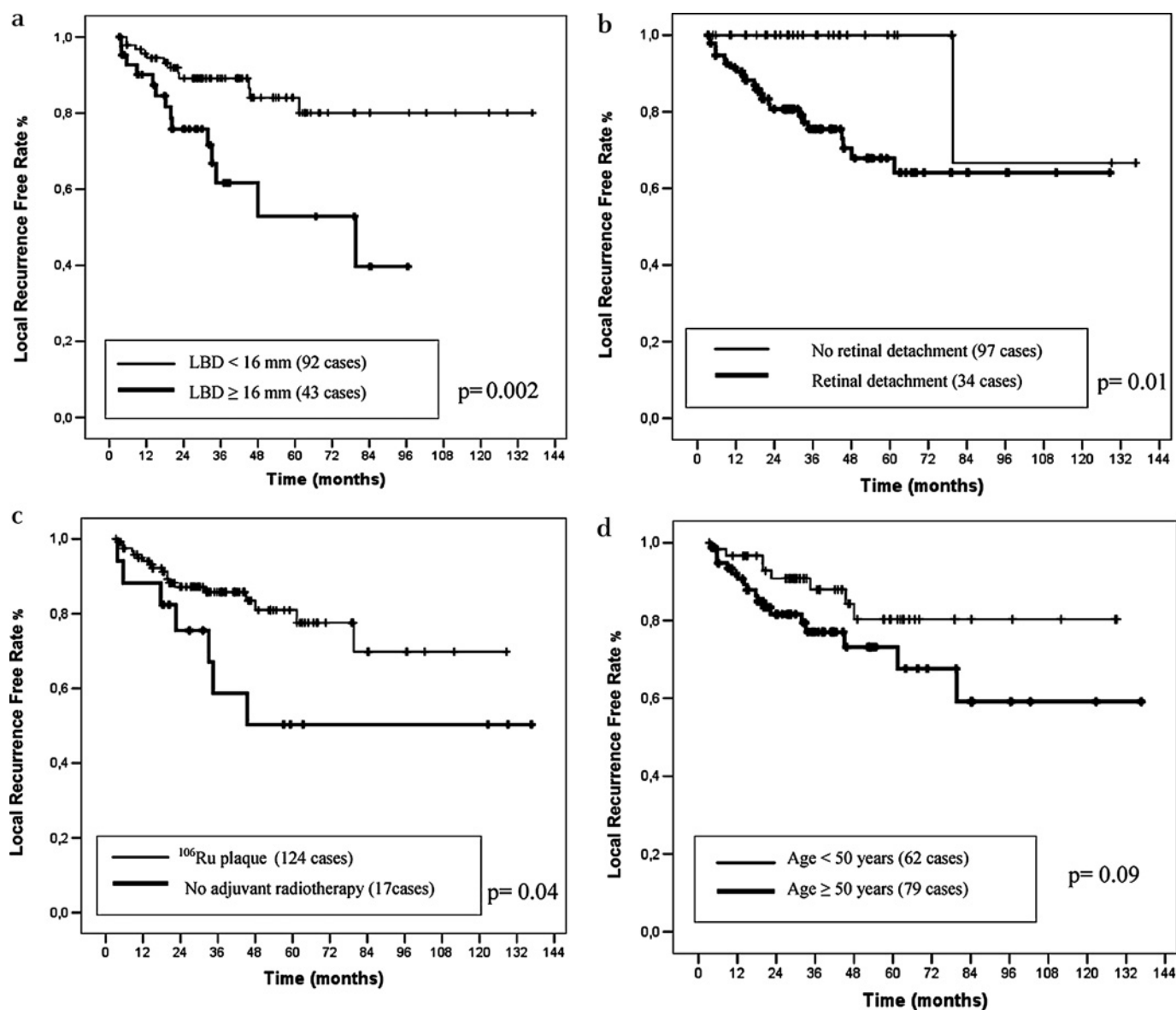


Figure 3 Kaplan–Meier curve showing the estimated recurrence-free rate for: (a) largest basal tumour diameter (LBD) <16 vs ≥16 mm; (b) absence of retinal detachment vs presence of retinal detachment; (c) ¹⁰⁶Ru plaque brachytherapy versus no adjuvant radiotherapy; and (d) age <50 vs ≥50 years.

Surgical techniques

The surgical technique used in this series was similar to the lamellar scleral dissection and choroidectomy under systemic arterial hypotension described by Foulds.^{13 14} After resection a pure SF₆ gas bubble (0.5–1.0 ml) was injected in the vitreous cavity and the adjuvant radiotherapy group had a 20.0 mm ¹⁰⁶Ru plaque (CCB; BEBIG, Berlin, Germany) placed on the excision bed. The prescription point for plaque dosimetry was defined in all cases as the apex of an imaginary 5.0-mm thick tumour receiving 100 Gy. The mean scleral dose was 470 Gy, ranging between 400 and 500 Gy. Residual tumours that did not regress spontaneously and recurrent tumours were submitted to further treatment consisting of photocoagulation, transpupillary thermotherapy (TTT), additional ¹⁰⁶Ru plaque brachytherapy, enucleation, exenteration or external beam radiotherapy. Postoperative complications such as vitreous or subretinal haemorrhages, extended retinal detachments, secondary cataract, radiotherapy-induced proliferative retinopathies or phthisis were

treated conventionally by vitreoretinal surgery (buckling procedures and/or pars plana vitrectomy), phacoemulsification with intraocular lens implantation, photocoagulation or enucleation.

Postoperative assessment

Follow-up assessment was conducted after 1, 3, 6 months and then annually. It comprised the same steps as the preoperative examinations with additional liver ultrasound and function tests every 6 months. All pigmentations newly detected during follow-up and treated not earlier than 3 months postoperatively were defined as recurrent tumours. Those lesions that were suspected of malignancy and did not regress spontaneously within the first three postoperative months were regarded as residual tumours.

Statistical methods

Pearson's χ^2 test was used to compare categorical factors between residual tumour groups. The t test was used for

confirmatory analysis of numerical data, and the Mann–Whitney U test was performed when the normal distribution assumption was violated. Univariate survival analysis relating to tumour recurrence included Kaplan–Meier curves and the log-rank test. Cox proportional hazards analysis was used to identify independent prognostic factors in the multivariate setting. Tumour recurrence was modelled as a time-dependent covariate when being analysed with respect to metastasis.

RESULTS

Follow-up varied from 96 days to 12 years. Patients had a mean age of 50 (range 17–74) years. There was ciliary body infiltration in 108 patients. Histopathological examination identified 64 spindle cell melanomas, 64 mixed type melanomas with spindle cell predominance, 21 mixed type melanomas with epithelioid cell predominance and one epithelioid cell melanoma. Thirteen per cent (20/150) received no adjuvant radiotherapy, while the remaining 87% (130/150) were treated adjuvantly with ^{106}Ru plaque brachytherapy. Exudative retinal detachment was detected in 105 patients (77%) at diagnosis. The mean preoperative visual acuity was 0.4 (table 1).

Residual tumour

A residual tumour was diagnosed in nine cases. The residual tumours were treated with TTT ($n=3$), argon laser photocoagulation (in the pre-TTT era) ($n=2$) or ^{106}Ru plaque brachytherapy ($n=3$), and one eye was enucleated. Three of the eight salvaged eyes later developed tumour recurrence.

Recurrent tumour

Tumour recurrences were detected in 26 eyes, between 4 and 78 months after TSR: 18 intraocular (unifocal/multifocal), six episcleral and two orbital. Seventeen recurrences were treated conservatively: eight by ^{106}Ru brachytherapy, three by argon laser photocoagulation, two by TTT, three by ^{106}Ru plaque brachytherapy with TTT and one by orbital radiotherapy. Nine eyes were enucleated or exenterated: seven primarily, one after photocoagulation and one after secondary ^{106}Ru plaque brachytherapy. Enucleation was combined in two cases with orbital external beam radiotherapy. Survival analysis showed 5- and 10-year recurrence rates of 24% and 32%, respectively. (figure 2)

Tumour recurrence was associated with LBD ≥ 16 mm ($p=0.002$), retinal detachment ($p=0.01$) and the lack of ^{106}Ru plaque brachytherapy ($p=0.04$) (table 2, figure 3).

Predictive factors for recurrent tumour were: lack of ^{106}Ru brachytherapy ($p=0.001$; HR 4.5), preoperative retinal detachment ($p=0.046$; HR 7.8), LBD ($p=0.035$; HR 2.4) and age ≥ 50 years ($p=0.046$; HR 2.6) (table 2).

Metastasis

Five- and 10-year metastatic rates were 28% and 44%, respectively (figure 4). Metastasis was associated with extrascleral tumour extension ($p=0.000$), tumour thickness ($p=0.000$) and local tumour recurrence ($p=0.008$) (figure 5). Metastasis tended to be associated with the presence of epithelioid cells and with a larger LBD (table 3). Tumour thickness ($p=0.003$; HR 4.4), local tumour recurrence ($p=0.06$; HR 2.3) and extrascleral tumour extension ($p=0.003$; HR 5.3) were identified as independent prognostic factors for metastasis (table 3, figure 5).

DISCUSSION

The present study found 5- and 10-year recurrence rates of 24% and 32% after TSR, respectively. Damato *et al* reported a 4-year

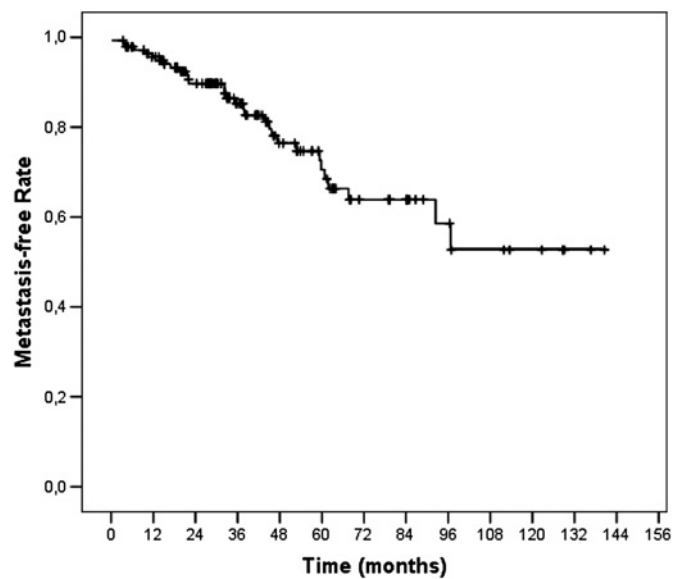


Figure 4 Kaplan–Meier curve showing the overall metastasis-free survival rate after transscleral resection (TSR).

recurrence rate that varied from 6% to 57%, depending on the identified risk factors.¹⁵ The risk factors reported by Damato *et al* were: epithelioid cellularity, posterior tumour extension, large tumour diameter and lack of ruthenium plaque radiotherapy. In 2003 Kivela *et al* reported in a case–control study a local recurrence rate of about 35% for TSR versus 5% for iodine brachytherapy for matched tumours averaging 8 mm in height, but only 15% of local recurrence if TSR was accompanied with adjuvant ruthenium brachytherapy.¹⁶ Their case selection of matched tumours averaging 8.0 mm in tumour height were very similar to the data set presented herein. We found a higher proportion of local tumour recurrences, which might be accounted for by the differences in surgical technique. We assume that our post-operative adjuvant ^{106}Ru brachytherapy, applied by a 20 mm plaque, is limited in its reduction of recurrences, especially in tumours with a tumour base >18 mm. In the present study brachytherapy was applied eccentrically and twice sequentially if the tumour base exceeded the irradiation field of the 20 mm plaque. Damato has suggested the placement of a 25 mm ^{106}Ru plaque, adding more safety in local tumour control.¹⁷ In addition, in order to address the possibility of inadequate excision margins, we chose to deliver 100 Gy to an imaginary residual tumour of 5.0 mm, in contrast to Damato who prescribed 100 Gy to a depth 1.0–2.0 mm with a relatively larger plaque.¹⁷ This resulted in our series in a mean scleral contact dose of 470 Gy, possibly resulting in some additional treatment related ocular morbidity (eg, dry eye symptoms, radiation retinopathy, etc.), which, however, was not the subject of the current analysis.

In 2007, Puusaari *et al* reported a 5-year local recurrence rate of 41% after transscleral local resection but only 7% after iodine brachytherapy in melanomas classified as large by the COMS criteria despite extension of tumours closer to the optic nerve and more frequent ciliary body involvement in the latter group.¹⁸ In study of Puusaari *et al*, the authors expressed concern about the increased risk of local tumour recurrence after TSR in large uveal melanomas. To further address the problem of tumour recurrence after TSR, postoperative adjuvant ^{106}Ru brachytherapy could be replaced by a form of preoperative ‘neoadjuvant’ radiotherapy covering the entire target volume, as has been proposed for endoresection.¹⁹ From 2004 onwards, we have performed

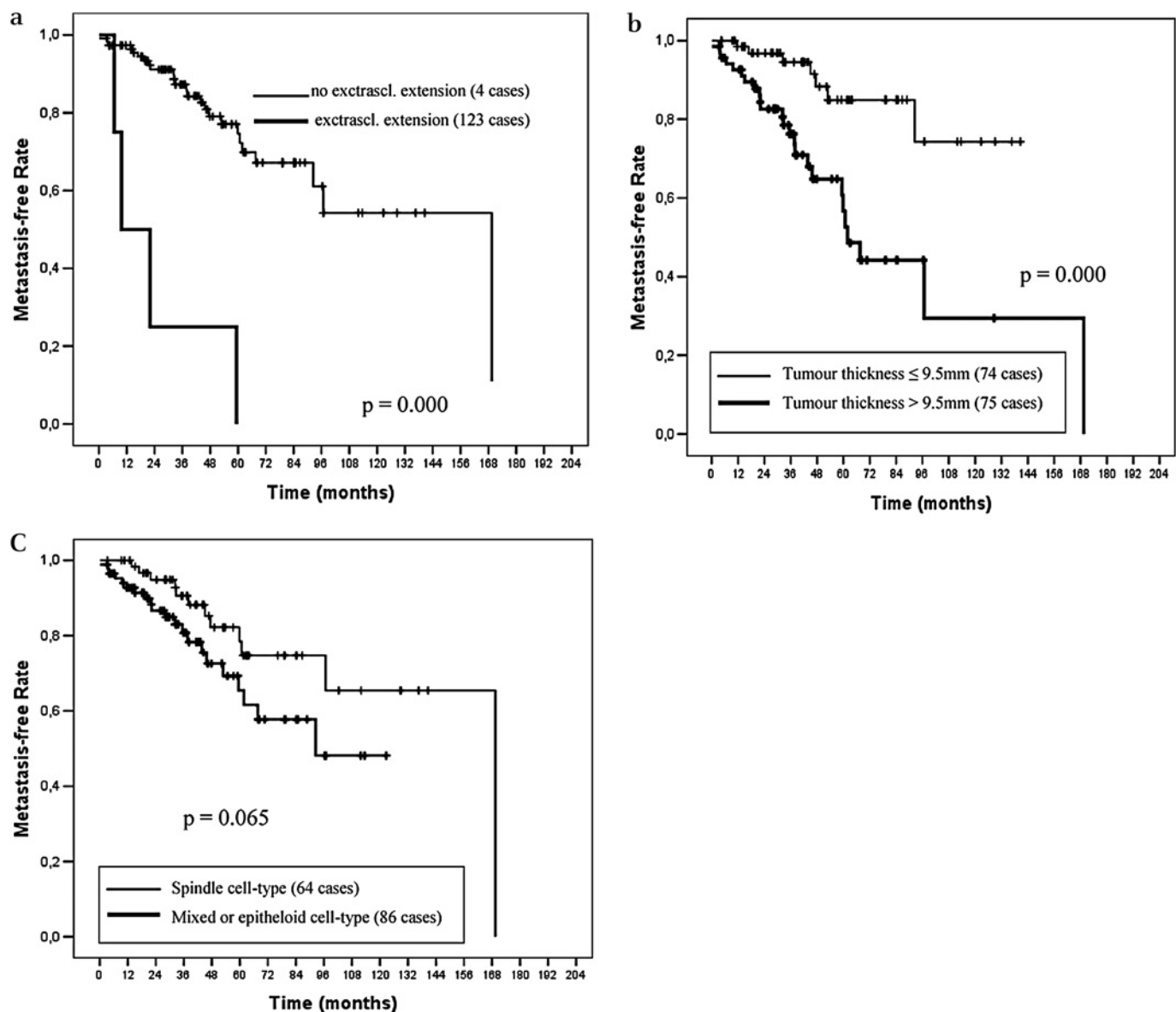


Figure 5 Kaplan–Meier curve showing the estimated metastasis-free rate for: (a) extrascleral tumour extension vs no extension; (b) tumour thickness ≤9.5 vs >9.5 mm; and (c) spindle cell type versus mixed or epithelioid cell type.

preoperative proton beam radiotherapy instead of 'postexcisional' adjuvant ^{106}Ru brachytherapy after TSR. These cases will be analysed at a later stage, regarding ocular and periocular morbidity, as well as tumour-related mortality, when longer-term follow-up data have been obtained in a larger number of patients.

In another interventional case series of 352 patients treated with plaque radiotherapy (^{106}Ru or ^{125}I) for uveal melanomas at least 8 mm thick, Shields *et al* found 5- and 10-year recurrence rates of 9% and 13%, respectively.²⁰ In their study, risk factors for tumour recurrence included ^{106}Ru plaque radiotherapy and ciliary body involvement.

In the search for risk factors associated with tumour recurrence after TSR, we found that the risk was 2.4 times higher with LBD ≥ 16 mm, 4.5 times higher without adjuvant ^{106}Ru brachytherapy and 7.8 times higher with retinal detachment. Retinal detachment was not previously identified as a risk factor for tumour recurrence; it could be attributable to the dissemination of melanoma cells in subretinal fluid distant from the tumour margins and to the correlation with very large tumours

as a confounding factor. Damato *et al* identified the following risk factors for local recurrences: a large LBD, lack of adjuvant radiotherapy, posterior tumour extension and epithelioid cell type.¹⁵

Only nine eyes had residual tumour and these were not associated with any specific tumour features.

The 5- and 10-year metastatic rates of 28% and 44%, respectively, in the present study are consistent with these published by others. Shields *et al* refer to 5- and 10-year metastatic rates of 30% and 55%, respectively, after plaque radiotherapy of tumours thicker than 8 mm.²⁰ The COMS reported a 10-year metastatic rate of 41% after primary enucleation.⁴ It did not identify local tumour recurrence as a risk factor for metastatic disease, probably because the COMS did not have enough patients with local tumour recurrence in the enucleation and iodine brachytherapy treatment arms. In the present study we identified extrascleral tumour extension at the time of diagnosis ($p=0.000$), tumour thickness ($p=0.000$) and local tumour recurrence ($p=0.008$) as independent predictive factors for the development of melanoma-related metastasis. p Values ranging from 0.06 to 0.09

Table 3 Metastasis after transscleral resection

				Multiple Cox regression analysis after forward and backward analysis		
Univariate analysis (log-rank test)						
Variable	Category	No. of patients with metastasis/ total no. of patients	p Value	HR	95% CI for HR	p Value
Sex	Female	16/77	0.74	4.5	1.7 to 12.3	0.003
	Male	17/73				
Age (years)	<50	14/66	0.31			
	≥50	19/84				
Ciliary body infiltration	Yes	24/108	0.81			
	No	9/42				
Tumour thickness (mm)	≤9.5*	7/74	0.000			
	>9.5	26/75				
LBD (mm)	<16*	19/97	0.09		0.4–2.5	0.9
	≥16	14/46				
TDO (mm)	<10	14/76	0.65			
	≥10	17/66				
TDF (mm)†	<10	13/60	0.09			
	≥10	7/47				
¹⁰⁶ Ru brachytherapy	Yes	27/130	0.45			
	No	6/20				
Histological type	Spindle cell	12/64	0.07	1.1	0.5–2.5	0.84
	Mixed or Epithelioid cell	21/86				
Tumour recurrence	Yes	15/26	0.008‡	2.3	1.1 to 4.9	0.04
	No*	18/115				
Extrascleral tumour extension	Yes	4/4	0.000	5.4	1.8 to 16.7	0.003
	No*	26/123				

*Reference category in Cox regression analysis.

†Multiple analysis not performed due to large number of missing values.

‡pValue according to Cox regression with time dependent variable.

LBD, largest tumour basal diameter; TDF, tumour distance to fovea; TDO, tumour distance to optic disc.

indicated a tendency for metastasis to be associated with epithelioid or mixed cell histology and large LBD.

The 6-year metastatic rate reported by Damato was 24% with two risk factors and 43% with three.¹⁵ He identified older age, presence of epithelioid cells, superior tumour location, LBD >16 mm, lack of adjuvant radiotherapy and secondary enucleation as risk factors for metastasis. While the association between an increased metastatic rate and extrascleral tumour extension is well known, tumour thickness was not unanimously identified as a risk factor for metastasis in the past. The fact that LBD was not significantly associated with metastasis could be due to the selection bias in this group, which consisted of the larger tumour spectrum having a mean LBD of 14.5 mm.

Better local tumour control is highly desirable, considering that tumour recurrence obviously influences the probability of metastasis. This has been published previously on patients treated by proton beam irradiation and is supported by our findings.²¹ Based on this rationale we have been performing preoperative proton beam radiotherapy instead of postoperative adjuvant ¹⁰⁶Ru plaque brachytherapy since 2004 in an attempt to reduce the local recurrence rate.

In summary, TSR is an alternative to enucleation in selected patients with large uveal melanomas eligible for this operation. This approach enables a significant number of candidates for enucleation to keep their affected eye, even though many will require additional surgery for postoperative complications.¹⁷ Adjuvant radiotherapy of the tumour area is strongly recommended to reduce tumour recurrences.^{17–22} Further improvement of tumour control may be achieved by using a larger ¹⁰⁶Ru plaque or possibly by implementing preoperative beam radiotherapy with a homogeneous tumour dose distribution (eg, proton beam teletherapy), sparing susceptible eye structures such as the optic

disc and macula. Whether this strategy will reduce ocular morbidity and, even more importantly, have an impact on patient survival remains to be elucidated by further studies.

Contributors We herewith state that all authors listed have contributed significantly in the conception and design, collection of data, as well as in the analysis and interpretation of data. All authors have drafted the article critically and have approved the version to be published.

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